# $\mathsf{D}A$ as a limit of the $\mathsf{D}_N A$

strict limit, homotopy limit, and a tower without fibrations

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**Abstract**

For a type $A$ of the poset model, $\mathsf{D}_N A$ is the localisation $\mathsf{D}A$ with the families indexed only by the simplices of dimension at most $N$. We show three facts. $\mathsf{D}A$ is the strict limit of the tower $(\mathsf{D}_N A)_N$. If $A$ is fibrant, $\mathsf{D}A$ is also its homotopy limit: the comparison map from $\mathsf{D}A$ to the internal limit, a type built with $\Pi$, $\Sigma$ and Path, is a homotopy equivalence. Finally, the maps $\mathsf{D}_{N+1}A \to \mathsf{D}_N A$ need not be fibrations, even when $A$ is fibrant, so the second fact is not a consequence of the first. All three are checked in `seg/core`.

## 1 Setting

**The poset model.** $\mathcal{C}$ is the category of finite non-empty posets and monotone maps, and $\Delta \subseteq \mathcal{C}$ the full subcategory of the linear posets $[n] = \{0 < \cdots < n\}$. A map of $\Delta$ is *strict* if it is strictly monotone, and *surjective* if it is onto. Every map $g$ of $\Delta$ factors uniquely as $g = c\sigma$ with $c$ strict and $\sigma$ surjective (Eilenberg–Zilber), and the same holds for a map $g : [k] \to P$ into a poset, with $c : [r] \to P$ injective. A context $\Gamma$ is a presheaf on $\mathcal{C}$, and a type $A$ over $\Gamma$ a presheaf on its category of elements; we write $A(X, \rho)$ for the fibre at $\rho \in \Gamma(X)$ and $a\,g$ for restriction. Every poset is identified with its nerve, so that $[m] \times [1]^k$ is $\Delta[m] \times \Delta[1]^k$. We write $i^*A$ for the restriction of $A$ to $\Delta$, a simplicial set over $i^*\Gamma$, and $\partial[m]$ for the union of the faces of $[m]$.

A *path* in $A$ at $(X, \rho)$ from $a$ to $b$ is an element of $A$ at $X \times [1]$ over $\rho\pi_1$ whose restrictions to $X \times \{0\}$ and $X \times \{1\}$ are $a$ and $b$. A family of paths natural in $X$ is a section of the type $\mathrm{Path}_A$. A *horn filling operation* on a simplicial set $T$ chooses fillers of horns $\Lambda^k[n] \to T$ such that the filler of the restriction of a horn along a face map is the restriction of the filler, whenever both are horns. We use two facts about fibrant types of the poset model.

(F1) If $A$ is fibrant, $i^*A$ has a horn filling operation.

(F2) If $i^*B$ has a horn filling operation, the type $\mathsf{D}B$ defined below is fibrant, naturally in $\Gamma$.

**Definition 1.1.** Let $A$ be a type over $\Gamma$. An element of $\mathsf{D}A$ at $(X, \rho)$ is a family

$$u(f) \in A([k], \rho f), \qquad f : [k] \to X,$$

with $u(f)\,g = u(fg)$ for every strict $g : [j] \to [k]$. Restriction along $X' \to X$ is precomposition. For $N \geq -1$, an element of $\mathsf{D}_N A$ at $(X, \rho)$ is the same with only the maps $f : [k] \to X$ with $k \leq N$. The map $r : \mathsf{D}_{N+1}A \to \mathsf{D}_N A$ forgets the values in dimension $N + 1$, and $v_N : \mathsf{D}A \to \mathsf{D}_N A$ forgets the values above $N$.

The important point is that the elements of $\mathsf{D}A$ and of $\mathsf{D}_N A$ are natural along the strict maps only. Their values on degenerate simplices are not determined by the values on the non-degenerate ones.

**The coskeleton.** An element of $K_N A$ at $(X, \rho)$ is a family $c(f) \in A([k], \rho f)$, for $k \leq N$ and $f : [k] \to X$, with $c(f)\, g = c(fg)$ for every $g : [j] \to [k]$ with $j \leq N$. So $i^* K_N A$ is the $N$-coskeleton of $i^* A$, and $\eta(a)(f) = af$ defines $\eta : A \to K_N A$.

**Lemma 1.2.**

1. *For $m \leq N$, $\eta : A([m], \tau) \to K_N A([m], \tau)$ is a bijection, with inverse $c \mapsto c(\mathrm{id})$.*
2. *If $i^* A$ has a horn filling operation, so has $i^* K_N A$.*

*Proof.* (1) The identity of $[m]$ is one of the indices, and $c(g) = c(\mathrm{id})\, g$.

(2) Let $h$ be a horn of $i^* K_N A$ on $\Lambda^k[m+1]$. If $m \leq N$, its faces are, by (1), elements of $A([m])$ forming a horn of $i^* A$; its filler $a$ gives the filler $\eta(a)$. If $m > N$, a map $[j] \to [m+1]$ with $j \leq N$ misses at least two vertices, hence a vertex other than $k$, so it lies in the horn, and the filler is $h$ read at these maps. Both cases commute with restriction to faces. □

**Proposition 1.3.** $\mathsf{D}_N A = \mathsf{D}(K_N A)$. *In particular $\mathsf{D}_N A$ is fibrant when $A$ is.*

*Proof.* Given $u \in \mathsf{D}(K_N A)(X, \rho)$, put $w(f) = u(f)(\mathrm{id})$ for $f : [k] \to X$, $k \leq N$; it is natural along strict maps. Conversely, given $w \in \mathsf{D}_N A(X, \rho)$, put $u(f)(g) = w(fc)\, \sigma$, where $g = c\sigma$ is the factorisation of $g : [j] \to [k]$ with $c$ strict and $\sigma$ surjective. Two such factorisations are related by a strict map, which gives the naturality of $u(f)$ along every map and of $u$ along strict maps. Below $N$ the value $w(fc)\sigma$ is $w(f)\, c\sigma = w(f)g$, so the two maps are inverse. The second statement follows from (F1), Lemma 1.2(2) and (F2). □

## 2 The strict limit

**Proposition 2.1.** $\mathsf{D}A$ *is the limit of the tower $\cdots \to \mathsf{D}_{N+1} A \xrightarrow{r} \mathsf{D}_N A \to \cdots \to \mathsf{D}_{-1} A = 1$, with the cone $(v_N)_N$.*

*Proof.* A compatible sequence $(u_N)_N$, $u_N \in \mathsf{D}_N A(X, \rho)$, defines the family $u(f) = u_k(f)$ for $f : [k] \to X$. Compatibility gives $u_k(f) = u_N(f)$ for $k \leq N$, so $u$ is natural along every strict map, since $u_N$ is for $N$ large. This $u$ is the unique element of $\mathsf{D}A(X, \rho)$ with $v_N(u) = u_N$ for all $N$, and the construction commutes with restriction. □

## 3 The maps of the tower are not fibrations

**Proposition 3.1.** *There is a fibrant type $A$ such that $r : \mathsf{D}_1 A \to \mathsf{D}_0 A$ has no fibration structure.*

*Proof.* Let $A$ be the constant type with value $\{0, 1\}$ over the empty context, which is fibrant. Then $A([k]) = \{0, 1\}$ for every $k$, and every restriction is the identity. An element of $\mathsf{D}_0 A$ at $X$ is an arbitrary function from the points of $X$ to $\{0, 1\}$, since the only strict map $[0] \to [0]$ is the identity. An element of $\mathsf{D}_1 A$ at $X$ is such a function together with a value on each map $[1] \to X$. Its value on an edge $e$ restricts, along the two strict maps $[0] \to [1]$, to its values on the two ends of $e$. So the ends of every edge get the same value.

Take $X = [0]$. The element $w = (0)$ of $\mathsf{D}_1 A([0])$ lies over $a = 0$. The element $\ell$ of $\mathsf{D}_0 A([1])$ with $\ell(0) = 0$ and $\ell(1) = 1$ is a line of $\mathsf{D}_0 A$ from $r(w)$ to 1. A lift of $\ell$ starting at $w$ is an element of $\mathsf{D}_1 A([1])$ over $\ell$. It must give a value to the identity of $[1]$, an edge from 0 to 1, and there is none. So $r$ does not even have the path lifting property. A fibration structure gives fillers for all open boxes over $[0] \times [1]$, and in particular such a lift. □

*Remark* 3.2. The same argument applies to $r : \mathsf{D}_{N+1}A \to \mathsf{D}_N A$ for every $N \geq 0$, for a fibrant $A$ whose restriction to $\Delta$ has an $N$-sphere with no filler. Lifting a line of $\mathsf{D}_N A$ over $[N]$ asks for $(N+1)$-simplices of $i^*A$ with a prescribed boundary. This is the coskeletal tower of a Kan complex: it converges strictly although its maps are not Kan fibrations.

So the strict limit of Proposition 2.1 is not the limit of a tower of fibrations, and it need not compute the homotopy limit. This is the content of the next section.

# 4 The homotopy limit

The *internal limit* of the tower is the type

$$L = \Sigma\,(u : \textstyle\prod_N \mathsf{D}_N A)\ \prod_N \mathrm{Path}_{\mathsf{D}_N A}(r\,u_{N+1}, u_N),$$

where the products range over the constant type of natural numbers. The *comparison* $\varphi : \mathsf{D}A \to L$ sends $u$ to $((v_N u)_N, \text{the constant paths})$.

If $A$ is fibrant, so is $L$: each $\mathsf{D}_N A$ is fibrant (Proposition 1.3), and fibrant types are closed under Path, $\Sigma$ and products over the constant type of natural numbers, the latter componentwise.

**Theorem 4.1.** *If $A$ is fibrant, $\varphi$ is a homotopy equivalence: there is a map $\psi : L \to \mathsf{D}A$, natural in the context, with paths $\varphi\psi \simeq \mathrm{id}_L$ and $\psi\varphi \simeq \mathrm{id}_{\mathsf{D}A}$ in the model.*

The rest of this section proves the theorem. Write $C = i^*A$ and $C_N = i^*K_N A$, the $N$-coskeleton of $C$. We identify a poset with its nerve, so that $[m] \times [1]^k$ is $\Delta[m] \times \Delta[1]^k$.

## 4.1 Tools

**Lemma 4.2** (Boxes)**.** *Let $T$ be a simplicial set with a horn filling operation, $P$ a finite poset, $S \subseteq P$ a simplicial subset and $e \in \{0,1\}$. Every map $P \times \{e\} \cup S \times \Delta[1] \to T$ extends to $P \times \Delta[1]$, with a choice of extension compatible with restriction along injective maps $P' \to P$.*

*Proof. Prisms over a simplex.* A problem at $n$ is a map $b : [n] \times \{e\} \cup Q \times \Delta[1] \to T$ with $Q \subseteq [n]$; its restriction along $d_i$ is a problem at $n-1$. By recursion on $n$ we choose a filler $F(b)$ with $F(b)\,d_i = F(b\,d_i)$. If $Q$ contains the identity, $F(b) = b$. Otherwise the fillers $F(b\,d_i)$ agree on common faces, and $\partial[n] \times \Delta[1] \cup [n] \times \{e\}$ is given. The non-degenerate $(n+1)$-simplices of $[n] \times [1]$ are the shuffles $P_k = ((0,0), \ldots, (k,0), (k,1), \ldots, (n,1))$, $0 \leq k \leq n$. The face $d_k$ of $P_k$ is the face $d_k$ of $P_{k-1}$, the face $d_{k+1}$ of $P_k$ is the face $d_{k+1}$ of $P_{k+1}$, and the other faces lie in $\partial[n] \times \Delta[1]$, apart from $d_{n+1}P_n = [n] \times \{0\}$ and $d_0 P_0 = [n] \times \{1\}$. For $e = 0$ fill $P_n, \ldots, P_0$ in this order, each as the filler of a horn on $\Lambda^k[n+1]$; for $e = 1$ fill $P_0, \ldots, P_n$, each on $\Lambda^{k+1}[n+1]$.

*Prisms over a poset.* For an injective $c : [n] \to P$ let $b_c$ be the problem read along $c$, with $Q = c^*S$. Then $b_c\,d_i = b_{cd_i}$, so $F(b_c)\,\mu = F(b_{c\mu})$ for every injective $\mu$. A simplex $g : [W] \to P$ factors uniquely as $g = c\sigma$, and we put $F(g,j) = F(b_c)(\sigma, j)$. If $k : [W'] \to [W]$, the factorisation of $gk$ is $(c\mu)\sigma'$ with $\mu$ injective and $\mu\sigma' = \sigma k$, so $F(gk, jk) = F(g,j)\,k$. If $g \in S$ then $c \in S$, the identity is in $c^*S$, and $F(g,j)$ is the given value; the same holds at $j = e$. □

We apply the lemma with $P = [m] \times Y$ and $S = \partial[m] \times Y \cup [m] \times Q$, for a poset $Y$ and a simplicial subset $Q \subseteq Y$. The *lid* is the value at $e$, the *sides* are the values on $\partial[m] \times Y$, and the *end data* are the values on $[m] \times Q$. For the squares, $Y = [1]_t$ and $Q = \{0,1\}$. For the cubes, $Y = [1]_t \times [1]_v$ and $Q$ is the boundary of the square: four pieces at $t = 0$, $t = 1$, $v = 0$, $v = 1$, which must agree at the four corners. The fillers are uniform along the face maps, so no excluded middle is used.

**Readings.** Let $P$ be a finite poset and $\phi : P \to X$. An element $z$ of $\mathsf{D}A$ (or of $\mathsf{D}_N A$, if the simplices of $P$ have dimension at most $N$) at $X$ is *read* along $\phi$ as the map $P \to C$ given on an injective chain $c : [r] \to P$ by $z(\phi c)$, and extended to all simplices by the Eilenberg–Zilber factorisation $g = c\sigma$, with $c$ injective and $\sigma$ surjective. The reading is natural along all maps of $\Delta$, although $z$ is natural along strict maps only.

Let $u \in \mathsf{D}A(X)$ and $f : [m] \to X$, and put $U_f = u(f) \in C_m$. Reading $u$ along $f\pi_1 : [m] \times [k] \to X$ gives a map $R_f^{[k]}$. For $k = 0$ it is the simplex $U_f$. We write $R_f = R_f^{[1]}$, a prism with both ends $U_f$.

**Lemma 4.3.** *$R_f$ need not be the constant prism at $U_f$. There is a square $Q_f$ over $[m] \times [1]_s \times [1]_v$ with*

$$Q_f(s,0) = R_f(s), \qquad Q_f(1,v) = R_f(v), \qquad Q_f(0,v) = R_f(v), \qquad Q_f(s,1) = U_f.$$

*Readings commute with the restriction to faces: $R_f$, $Q_f$ and $U_f$ restricted along $d_i \times \mathrm{id}$ are $R_{fd_i}$, $Q_{fd_i}$ and $U_{fd_i}$.*

*Proof.* On the edge $(i,0) \to (i,1)$ of $[m] \times [1]$, $R_f$ is $u$ at a degenerate simplex of $X$. The naturality of $u$ along strict maps says nothing about this value, which is not determined by $u(f)$.

Let $\beta : [1] \times [1] \to [2]$ be the monotone map with $\beta(s,0) = s$, $\beta(1,v) = 1 + v$, $\beta(0,v) = 2v$ and $\beta(s,1) = 2$. Put $Q_f = R_f^{[2]} \circ (\mathrm{id} \times \beta)$. The edges $v = 0$, $s = 1$ and $s = 0$ of $\beta$ are the three strict maps $[1] \to [2]$. Reading along a strict inclusion $P' \to P$ is reading on $P'$, since the injective chains of $P'$ are injective chains of $P$. So these three edges of $Q_f$ are $R_f$. The edge $v = 1$ is $R_f^{[2]}$ precomposed with $[m] \to [m] \times \{2\}$, which is again a strict inclusion, so it is the reading of $u$ along $f$, which is $U_f$. Note that we precompose an already extended map. We do not claim that reading commutes with projections.

The last statement is the same argument for the strict inclusions $d_i \times \mathrm{id}$, together with $u(f)\, d_i = u(fd_i)$. □

## 4.2 The map $\psi$

Let $\ell = (u_N, p_N) \in L(X)$, so $p_N$ is an element of $\mathsf{D}_N A$ at $X \times [1]$ from $r\, u_{N+1}$ to $u_N$. For $f : [m] \to X$, let $\sigma_N(f)$ be the reading of $u_N$ along $f$ and $p_N(f)$ the reading of $p_N$ along $f \times [1]$. We build, by recursion on $m$ and, for each $f$, on the stage $N \geq m + 1$:

- an $m$-simplex $\psi(f)$ of $C$;
- prisms $H^N(f)$ over $[m] \times [1]_t$ from $\psi(f)$ $(t = 0)$ to $\sigma_N(f)$ $(t = 1)$;
- squares $\Theta_N(f)$ over $[m] \times [1]_t \times [1]_s$ with $\Theta_N(t,1) = H^N(t)$, $\Theta_N(t,0) = H^{N+1}(t)$, $\Theta_N(0,s) = \psi(f)$ and $\Theta_N(1,s) = p_N(f)(s)$.

Each of these restricts along $d_i$ to the same datum at $fd_i$.

1. $H^{m+1}(f)$ fills the box over $[m] \times [1]_t$ with lid $\sigma_{m+1}(f)$ at $t = 1$ and sides $H^{m+1}(fd_i)$. Its end at $t = 0$ is $\psi(f)$.
2. $\Theta_N(f)$ fills the box over $[m] \times [1]_t$, line $s$, with lid $H^N(f)$ at $s = 1$, end data $\psi(f)$ at $t = 0$ and $p_N(f)$ at $t = 1$, and sides $\Theta_N(fd_i)$. Its end at $s = 0$ is $H^{N+1}(f)$.

The face $fd_i$ has dimension $m - 1$, so its data exist for $N \geq m$, and in particular at the stages $N \geq m + 1$ used at $f$. All these boxes lie in dimension at most $m + 2 \leq N + 1$. They live in $C$: the stage only says which of $u_N$ and $p_N$ is read. Then $\psi(\ell) = (f \mapsto \psi(f))$ is natural along the face maps, hence along all strict maps, and is an element of $\mathsf{D}A(X)$.

The value of $\psi(\ell)$ in dimension $m$ is taken at stage $m+1$, never lower. In particular $u_0$ is never used, so the obstruction of Section 3 does not arise. The squares $\Theta_N$ are part of the data and are built by box filling, not by concatenating the paths $p_N$: a concatenation would not commute strictly with the faces.

### 4.3 The path $\varphi\psi \simeq \mathrm{id}_L$

A path in $L$ from $\varphi\psi(\ell)$ to $\ell$ consists of lines $l_N$ of $\mathsf{D}_N A$ from $v_N\psi(\ell)$ to $u_N$, and squares relating $r\, l_{N+1}$, $l_N$, the constant line and $p_N$. The line $l_N$ is read off $H^N(f)$ at the maps $(f,h) : [k] \to X\times[1]$, $k \le N$, through $\langle \mathrm{id}, h\rangle$: its value at $(f,h)$ is $H^N(f)$ at $(\mathrm{id}, h)$, natural along strict maps because $H^N$ is face compatible. For $k < N$ the prism $H^N(f)$ exists. For $k = N$ it is built in $C_N$: $H^N(f)$ and $\Theta_N(f)$ fill, in $C_N$, the box formed by $H^{N+1}(f)$, $p_N(f)$, the constant line, and the sides $\Theta_N(fd_i)$. The squares are read off $\Theta_N$ in the same way, and their edges are the required ones.

### 4.4 The path $\psi\varphi \simeq \mathrm{id}_{\mathsf{D}A}$

Let $u \in \mathsf{D}A(X)$ and run the construction on $\varphi(u)$. Its path $p_N$ is constant, so its reading $p_N(f)$ is $R_f$ (Lemma 4.3). Thus $H^N(f)$ goes from $\psi\varphi(u)(f)$ to $U_f$, and

$$\Theta_N(t,1) = H^N(t), \quad \Theta_N(t,0) = H^{N+1}(t), \quad \Theta_N(0,s) = \psi\varphi(u)(f), \quad \Theta_N(1,s) = R_f(s).$$

We build, by recursion on $m$ and on $N \ge m+1$, a prism $G(f)$ over $[m]\times[1]_t$ from $\psi\varphi(u)(f)$ to $U_f$ and squares $\Xi^N(f)$ over $[m]\times[1]_t\times[1]_v$ with

$$\Xi^N(t,0) = H^N(t), \quad \Xi^N(t,1) = G(t), \quad \Xi^N(0,v) = \psi\varphi(u)(f), \quad \Xi^N(1,v) = R_f(v),$$

all restricting along $d_i$ to the data at $fd_i$. We also keep the cubes filled on the way.

1. *Initial square.* $\Xi^{m+1}(f)$ fills the box over $[m]\times[1]_t$, line $v$, with lid $H^{m+1}(f)$ at $v=0$, end data $\psi\varphi(u)(f)$ at $t=0$ and $R_f(v)$ at $t=1$, and sides $\Xi^{m+1}(fd_i)$. Its end at $v=1$ is $G(f)$, which goes from $\psi\varphi(u)(f)$ to $R_f(1) = U_f$, with faces $G(fd_i)$.
2. *Successor cube.* $\Xi^{N+1}(f)$ is the end at $s=0$ of the filler of the box over $[m]\times[1]_t\times[1]_v$, line $s$, with lid $\Xi^N(f)$ at $s=1$, sides the cubes at $fd_i$, and end data

$$\begin{array}{ll} t=0: & \psi\varphi(u)(f), \\ t=1: & Q_f(s,v), \\ v=0: & \Theta_N(t,s), \\ v=1: & G(t), \text{ independent of } s. \end{array}$$

The pieces agree at the corners: at $t=1, v=0$ both are $R_f(s)$; at $t=1, v=1$ both are $U_f$; at $t=0$ all are $\psi\varphi(u)(f)$. At $s=1$ they are the edges of $\Xi^N(f)$: $Q_f(1,v) = R_f(v)$, $\Theta_N(t,1) = H^N(t)$. At $s=0$ they give the required edges of $\Xi^{N+1}(f)$: $Q_f(0,v) = R_f(v)$ and $\Theta_N(t,0) = H^{N+1}(t)$. On a face the end data are those of the face, by Lemma 4.3 and the face relations of $\psi$, $H$, $\Theta$ and $G$. The sides agree with the lid because the cube at $fd_i$ starts at $\Xi^N(fd_i)$, which is $\Xi^N(f)$ on that face. The sides agree with each other where two faces meet, by the face relations one dimension lower. All boxes are filled in $C$ itself: $R_f$ and $Q_f$ come from the full element $u$, so no truncation bound appears.

The prisms $G(f)$ are face compatible, so the values $G(f)(\mathrm{id}, h)$ for $(f,h) : [n] \to X \times [1]$ form an element of $\mathsf{D}A(X\times[1])$, natural along the strict maps, from $\psi\varphi(u)$ to $u$.

### 4.5 Naturality

The data at $f$ are computed from the readings of the input along $f$ and from the data at the faces of $f$. For $h : Y \to X$, the problem at $f$ for the restriction of the input along $h$ is the problem at $hf$ for the input. The fillers are chosen uniformly, so equal problems have equal fillers, by induction on the dimension and on the stage. For $\psi\varphi \simeq \mathrm{id}$ one needs in addition that the readings $R_f$ and $Q_f$ of the restriction $h^*u$ are those of $u$ at $hf$, and that the restriction of $\varphi(u)$ along $h$ is $\varphi(h^*u)$. So $\psi$ and both homotopies are natural in $X$. A natural family of such cylinders is a section of the path type, by the definition of paths in Section 1. This proves Theorem 4.1.

*Remark* 4.4. No choice principle is used. The connectedness of the maps of the tower is witnessed by box fillers of $C$ and of the $C_N$, chosen uniformly along the face maps. The internal argument from towers of connected maps would need dependent choice.

## A Correspondence with the formalisation

Layer L9 of `seg/core`, over the poset instance.

| statement | files |
|---|---|
| $\mathsf{D}_N A = \mathsf{D}(K_N A)$, fibrancy | CCosk, CCoskHorn, CLocDN, CLocDNC |
| Proposition 2.1 | CLocDN (Lim) |
| $L$, $\varphi$ | CPostLimit (Product, Tower, Cone, Post); $L$ fibrant: CPostFib |
| boxes | CBoxFill; over $[m] \times [1]$: CPostSq; over $[m] \times Y$: CPostBox; the boundary of the square and its four pieces: CPostCube (Geo, E4, E4Cmp) |
| readings, Lemma 4.3 | CPostMap (DR), CPostK; for $u \in \mathsf{D}A$: CPostRead ($\beta$, Rd, Sq: $R$, $R2$, $Q$ and its edges, Face) |
| $\psi$ | CPostRec, CPostStep0/1/3, CPostInput |
| $\varphi\psi \simeq \mathrm{id}$ | CPostStep2, CPostLine, CPostSquare, CPostAssemble; naturality CPostExt, CPostSqNat, CPostNat, CPostHomNat; the path: CPostConvergence (RightInverse.homotopy) |
| $\psi\varphi \simeq \mathrm{id}$ | CPostInvIn (the recursion on $\varphi(u)$), CPostInv ($G$, $\Xi$, the cubes), CPostInvAsm (the cylinder, its ends), CPostInvNat (naturality; LeftInverse.homotopy) |

Proposition 3.1 is not formalised. The fibrancy of $L$ is CPostFib (ProductFib: countable products; HTFib: the path types of the edges; TowerFib, PostFib: $L$).